

Text Mining Assignment - Report

Group 53

Book: The Travels of Marco Polo, Volume 1 (Project Gutenberg eBook #10636)

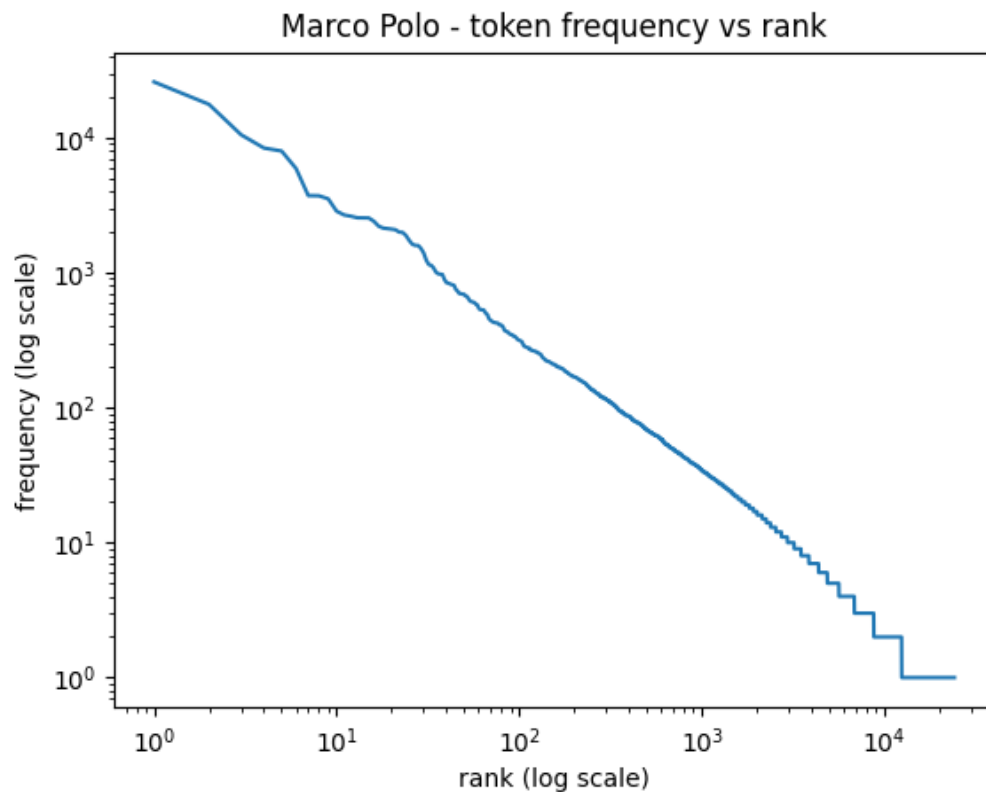
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(a) Book

The chosen book is **The Travels of Marco Polo, Volume 1 (Project Gutenberg eBook #10636)**.

(b) Token frequency vs rank (log-log)

The book was tokenized into lowercase word tokens (punctuation and numbers dropped) and each token counted. The plot below shows token frequency against rank on log-log axes - the near-straight line is the signature of Zipf's law.



(c) Top 20 tokens

top 20 tokens:

the 25900

of 17671

and 10510

in 8385

to 7964

a 5910

is 3724

that 3713

i 3524

by 2856

it 2676

as 2618

his 2550

was 2545

ii 2543

s 2405

on 2211

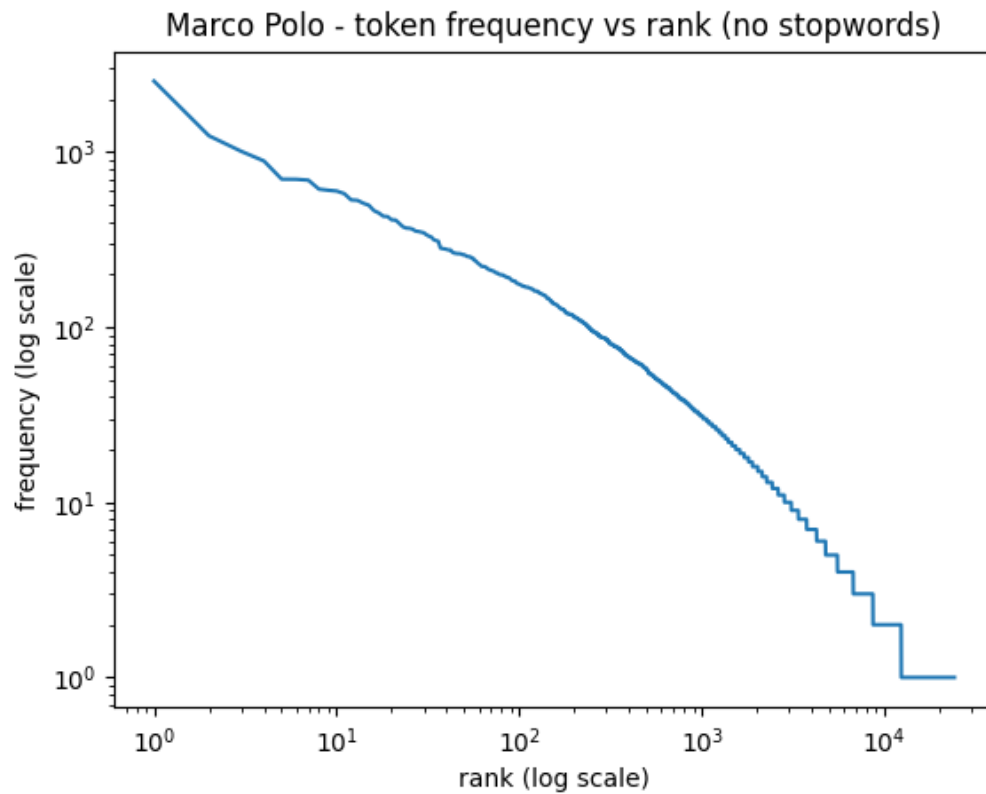
which 2139

with 2123

from 2102

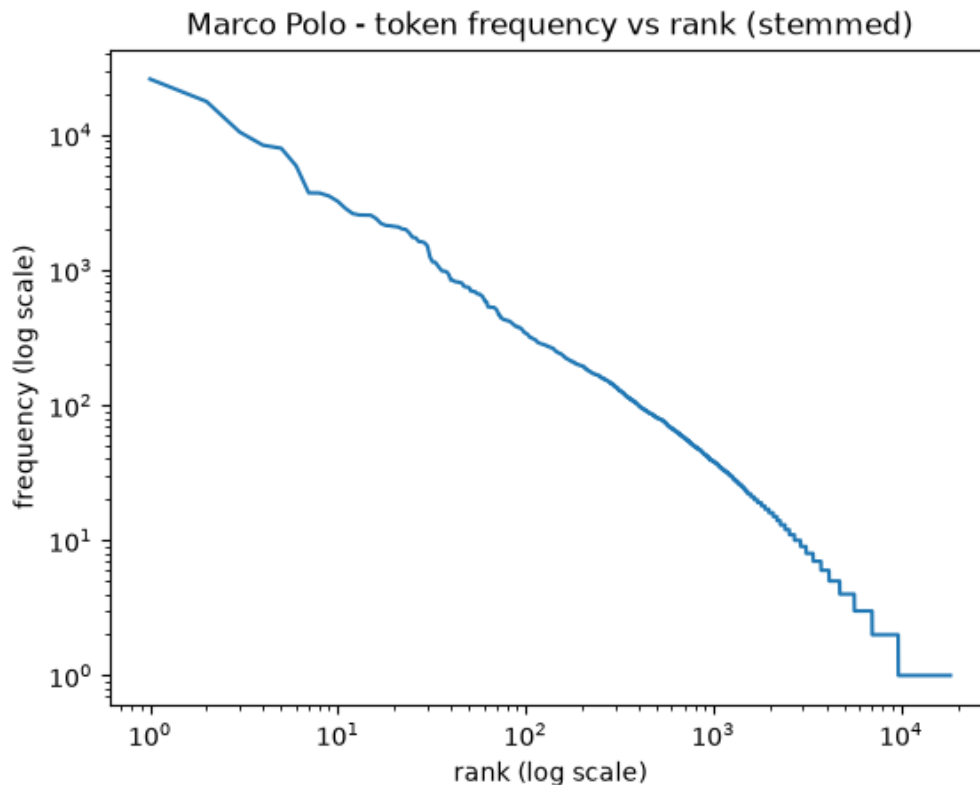
(d) Frequency vs rank without stopwords

The same analysis after removing English stopwords. Removing the closed set of very frequent function words strips the top of the ranking, leaving content words.



(e) Frequency vs rank with stemming

The tokens were reduced to their stems with the Porter stemmer, then counted and plotted on log-log axes.



Comparison (original / no-stopwords / stemmed):

Comparison of token counts across parts (b), (d) and (e):

Tokenizing the whole book gives 333590 tokens made up of 24226 unique word types (part b). Removing English stopwords barely changes the number of unique types (24084) because stopwords are a small closed set of very frequent words, but it removes many token occurrences (172803 tokens left), which is why the top of the frequency ranking changes the most. Stemming with the Porter stemmer keeps the total number of tokens the same as in (b) (333590) but collapses inflected forms such as "travel", "travels" and "travelling" onto a single stem, cutting the vocabulary sharply to 18303 unique stems. As a result the stemmed frequencies are higher per type and the log-log Zipf curve stays straight but shifts left, since fewer distinct types now carry the same total mass.

(f) A POS tagging mistake

POS tagging mistake (part f)

Sentence: "The remainder, however, is favourable to agriculture, and produces everything abundantly."

The word "agriculture" is tagged VB (base-form verb), but it is actually a common noun (NN): it is the object of the preposition "to" in the phrase "favourable to agriculture".

Why the tagger is fooled: the token "to" is by far most often the infinitive marker, which is normally followed by a base-form verb (e.g. "to run", "to go"). The averaged-perceptron tagger leans heavily on the previous tag, so after seeing "to" (tagged TO) it strongly prefers a following VB and overrides the fact that "agriculture" is only ever a noun. Here "to" is really a preposition, so the correct tag is NN.

(g) Proper-noun (NNP/NNPS) tag cloud

A tag cloud of the proper nouns in the book (case preserved). See the written explanation of whether it matches the book below.



The tag cloud of proper nouns (NNP/NNPS) matches the book very well. The most frequent proper nouns are Polo (820), Marco (650), Kaan (316), Khan (274) and Kublai (256) - the traveler himself and the Mongol emperor he served - together with the places on his route: China, India, Persia, Venice, Cathay, Hormuz and Tibet. Names like Yule, Pauthier and Ramusio also appear prominently because this Gutenberg edition includes extensive commentary by its translator Sir Henry Yule and refers to earlier editors, and those notes are part of the text. We filtered out all-caps tokens (from chapter headings like "THE TRAVELS OF...") and single letters (scholars' initials such as "H. Yule"), which the POS tagger mis-labels as proper nouns.

(h) Regex for two consecutive repeated words

regex: \b([a-zA-Z]+)[,.;:!?\"']*\\s+\\1\\b
number of matches: 81

it. it (x7)
had had (x5)
w. w (x4)
j. j (x4)
h. h (x3)
etc. etc (x3)
t. t (x2)
etc., etc (x2)
it, it (x2)
you, you (x2)
c., c (x1)
blanche; blanche (x1)
no, no (x1)
was, was (x1)
thousand thousand (x1)
alor! alor (x1)
lamba. lamba (x1)
tuez, tuez (x1)
derbend, derbend (x1)
tauris. tauris (x1)
yasdi. yasdi (x1)
kerman. kerman (x1)
r. r (x1)
efmerveillez, efmerveillez (x1)
walashjird. walashjird (x1)
it; it (x1)
is is (x1)
vous vous (x1)
balc. balc (x1)
badashan. badashan (x1)
keshimur. keshimur (x1)
samarcand. samarcand (x1)
yarcand. yarcand (x1)
cotan. cotan (x1)
pein. pein (x1)
charchan. charchan (x1)
camul. camul (x1)
chingintalas. chingintalas (x1)
campichu. campichu (x1)
caracoron. caracoron (x1)
v. v (x1)
slaughter! slaughter (x1)
musk; musk (x1)
all, all (x1)
is, is (x1)
that, that (x1)
thousands, thousands (x1)
there. there (x1)
long long (x1)
g. g (x1)
d. d (x1)
hili, hili (x1)
konkan, konkan (x1)
lewis lewis (x1)

phungan, phungan (x1)
sien, sien (x1)
i., i (x1)